

构造地貌学 课程汇报

综述：新生代贝加尔裂谷带的发育

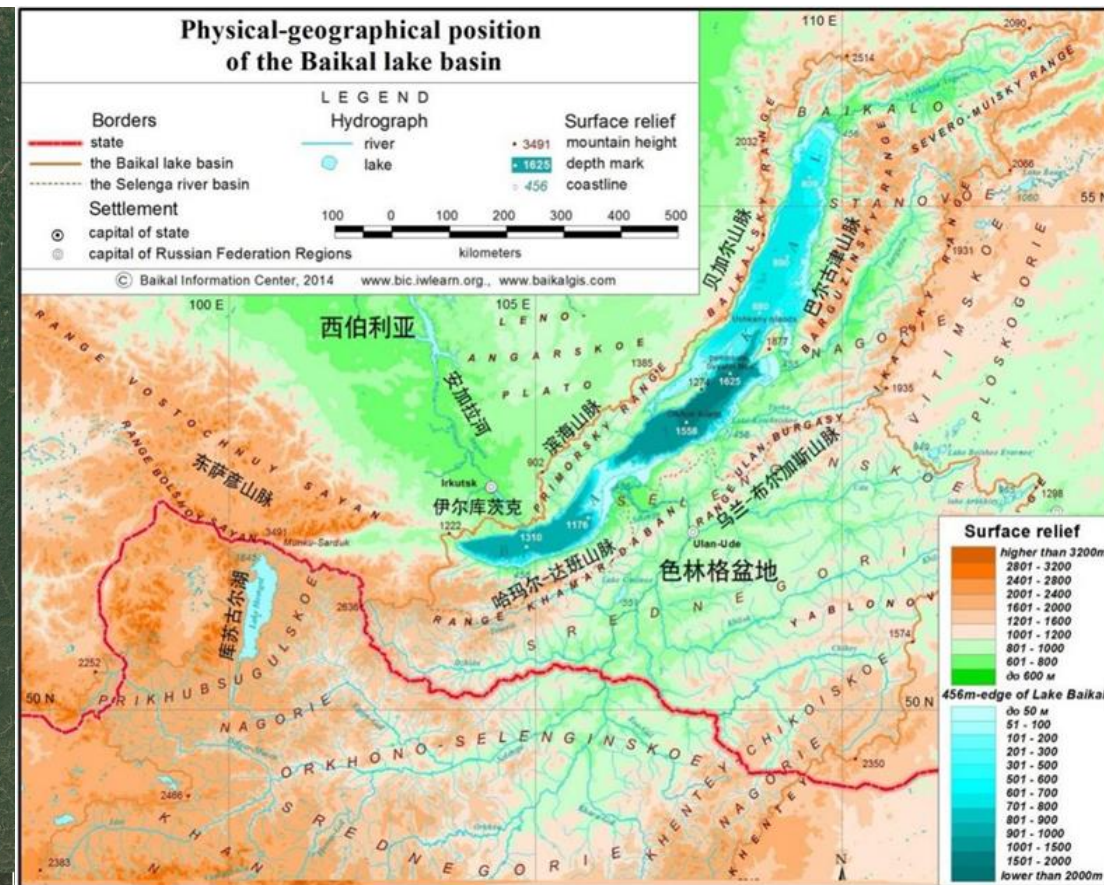
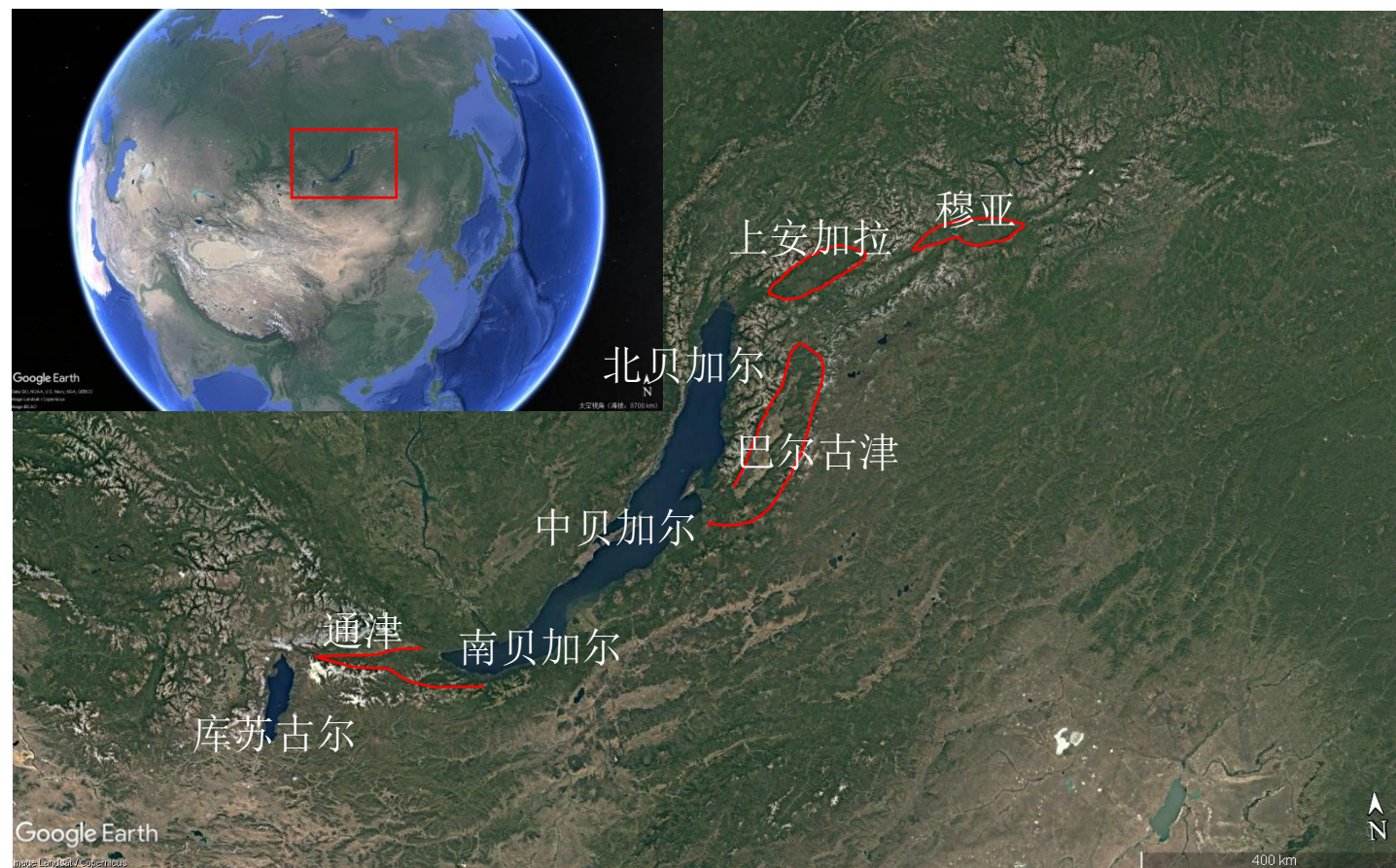
A Review: the Development of Baikal Rift Zone since Cenozoic

茅单文 SID: 231830128

2023级 地球科学与工程学院 行星科学

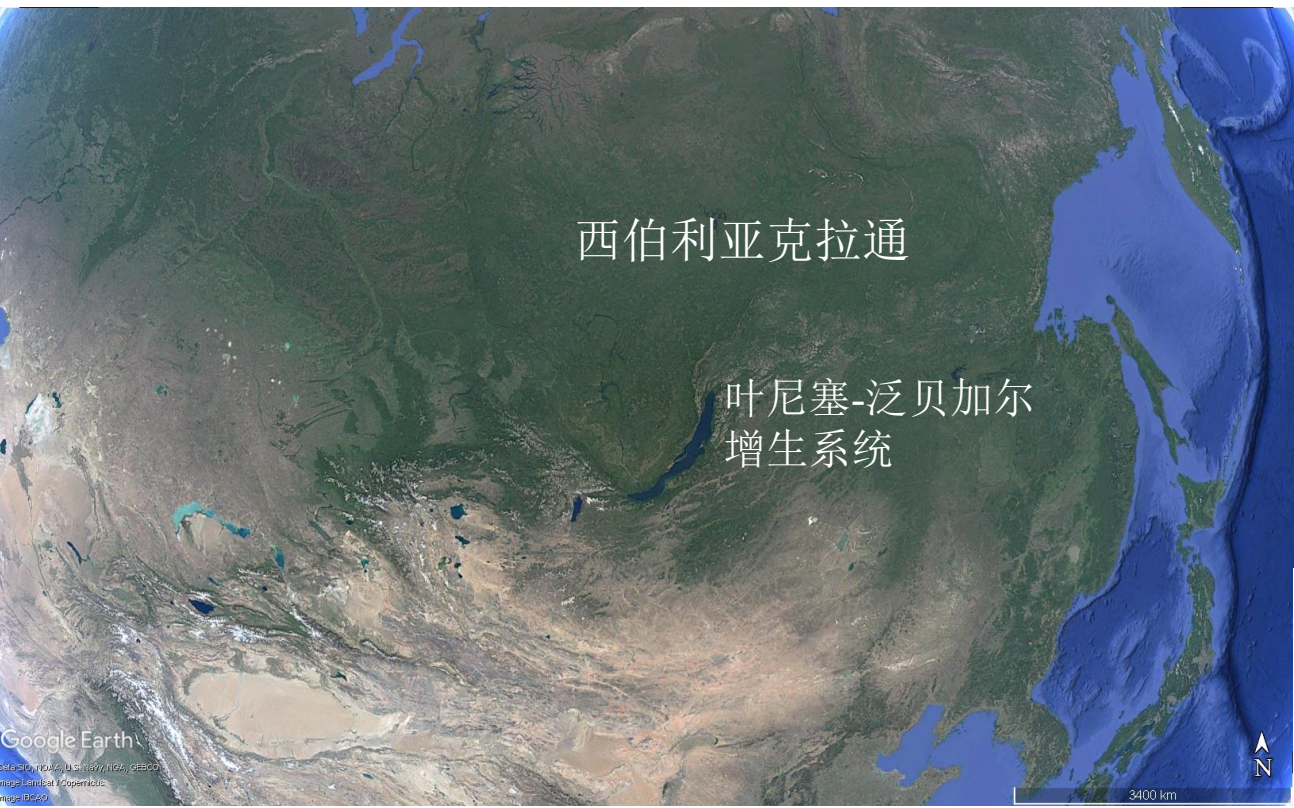
2025/06/04

区域地形

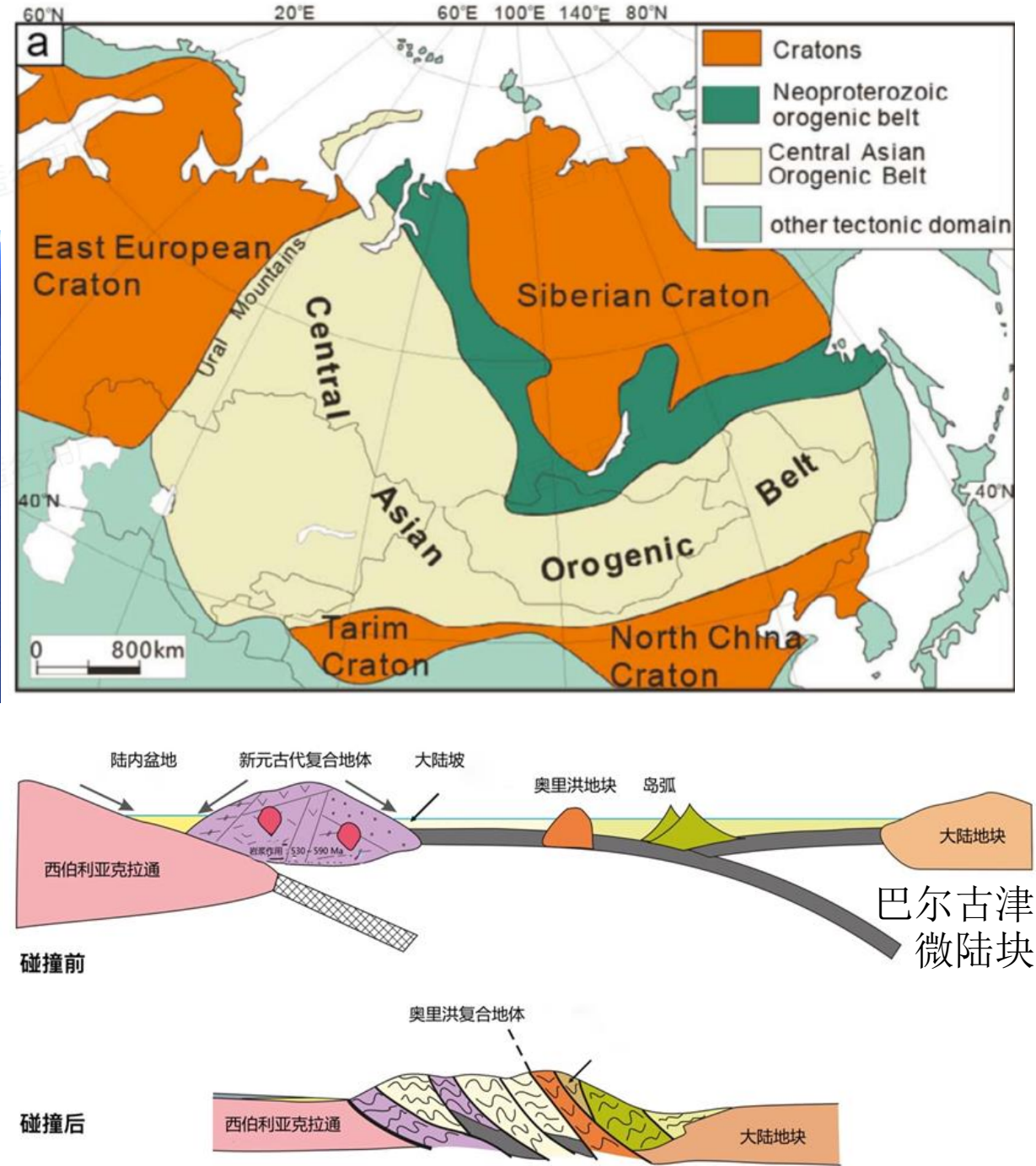


一系列裂谷盆地，部分充水成湖

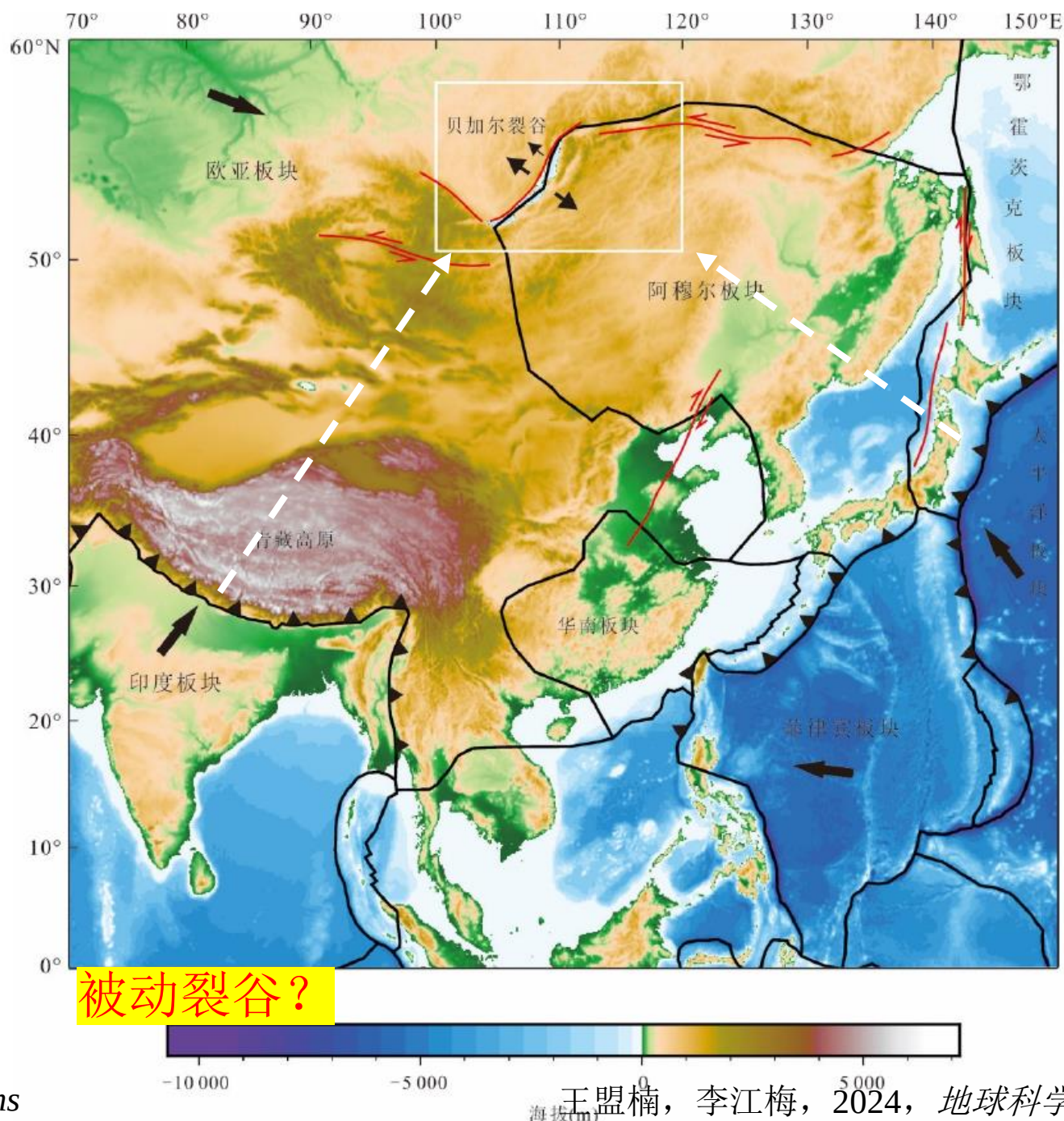
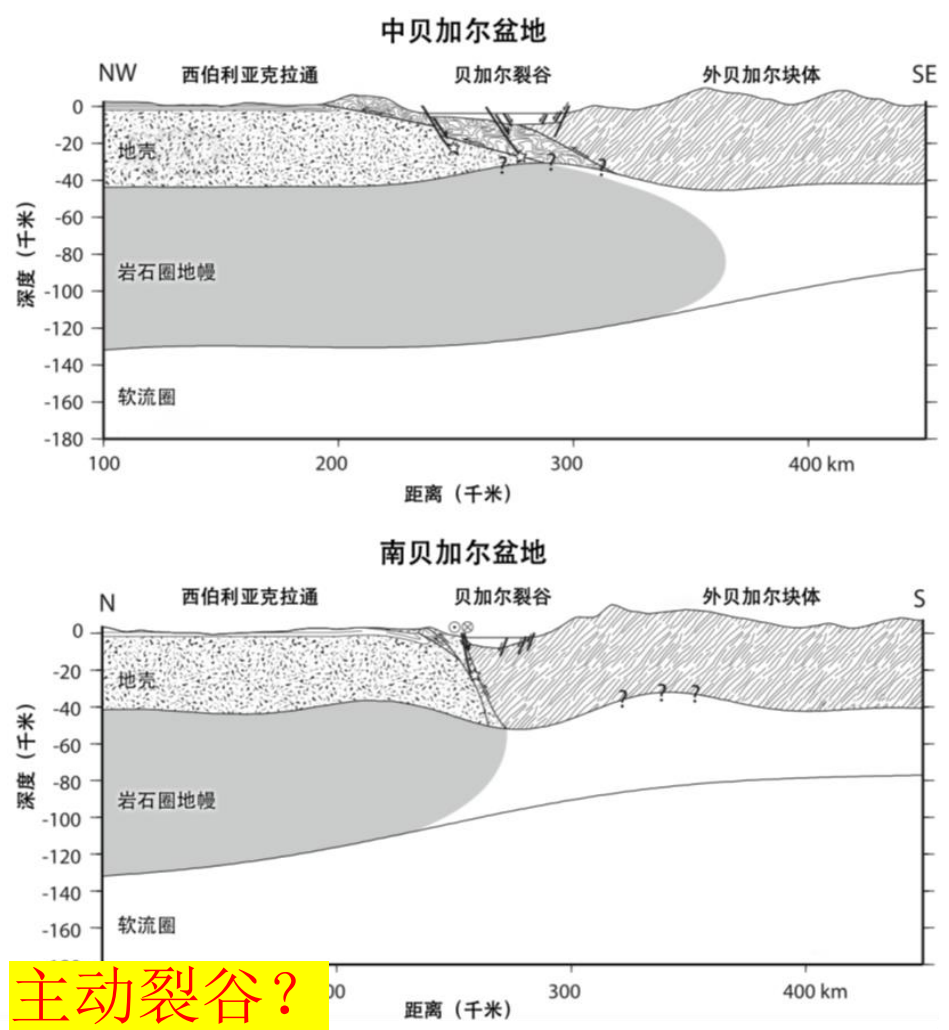
区域构造历史：早古生代



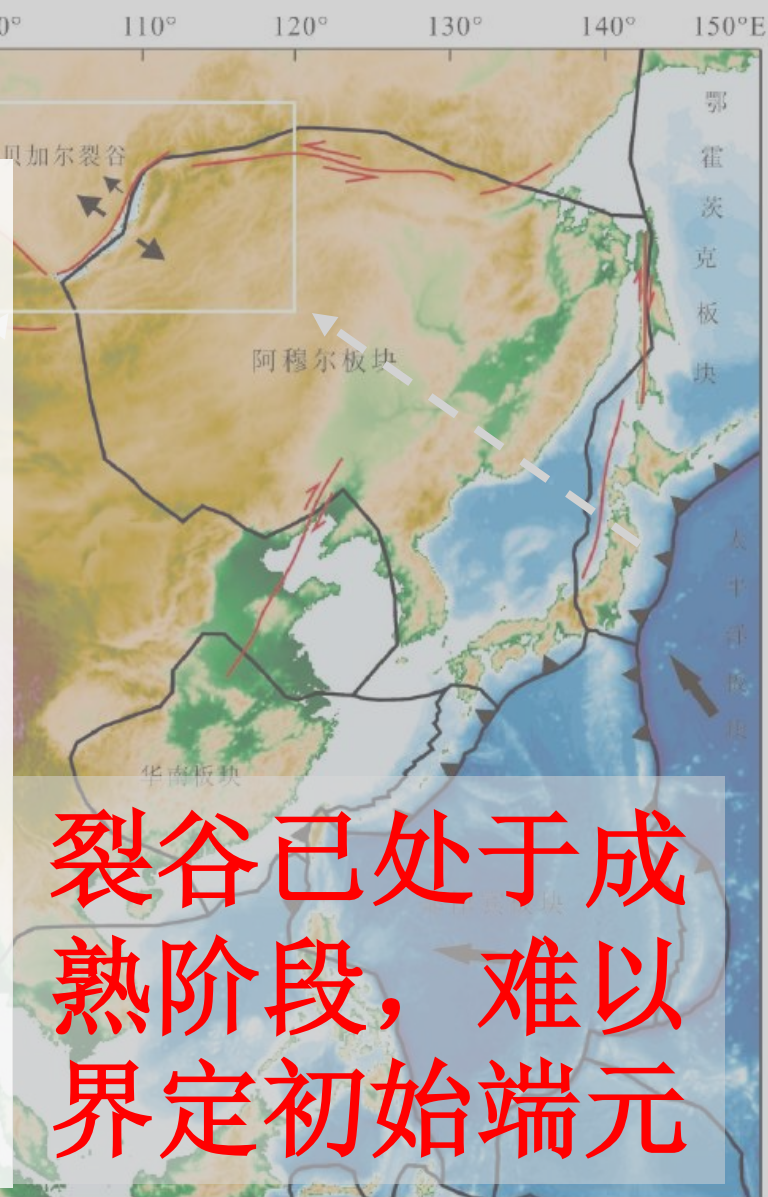
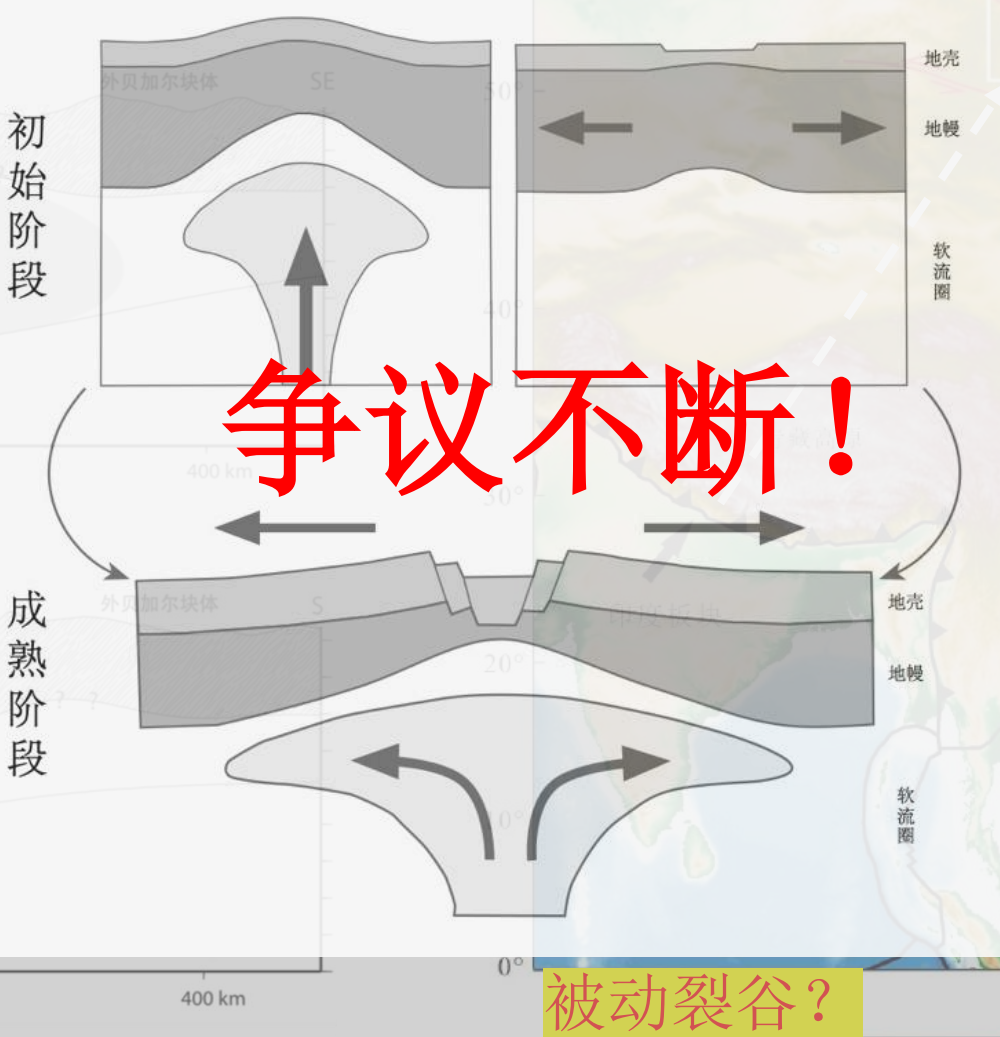
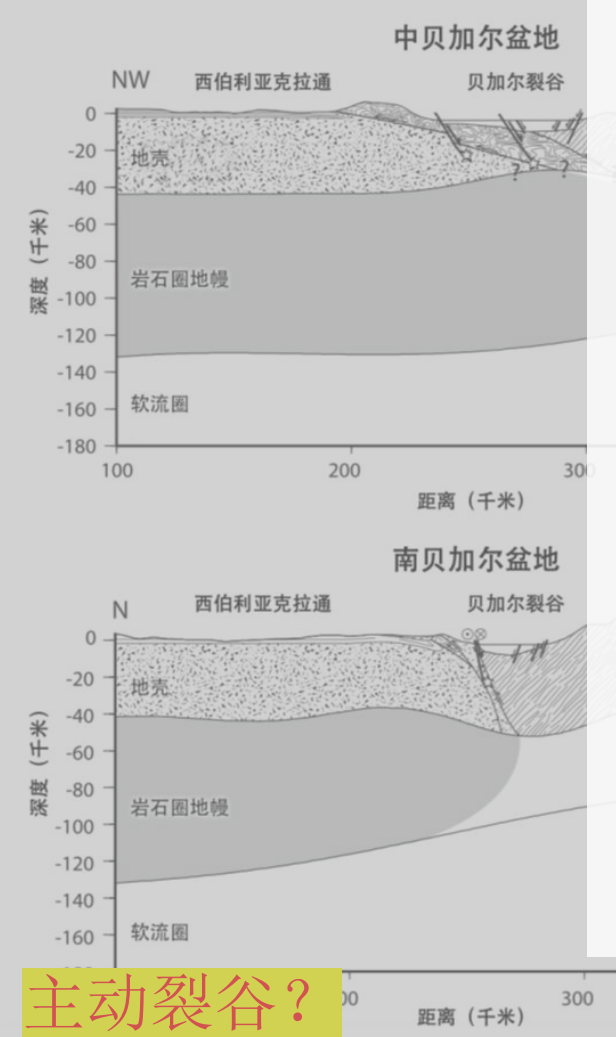
西伯利亚克拉通-巴尔古津微陆块 陆陆碰撞
后续板块拼贴 ⇨ 叶尼塞-泛贝加尔增生系统



区域构造历史：新生代



区域构造历史：新生代



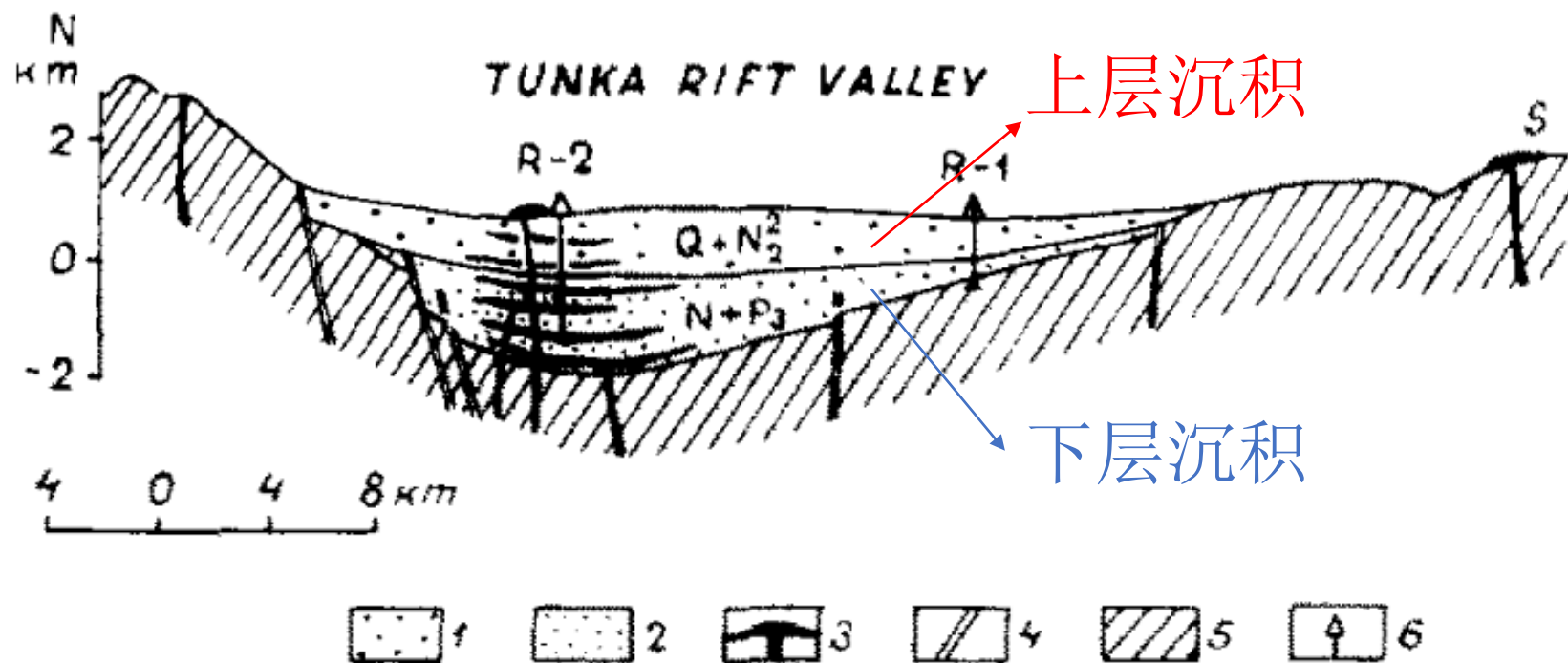
主动裂谷?

被动裂谷?

裂谷沉积地层：发育的两个阶段

晚上新世（3~4Ma）~至今

粗粒碎屑沉积
+ 洪积相、滑坡相、冰川相



分选性更差

渐新世（30~35Ma）~晚上新世（3~4Ma）

细粒碎屑沉积，碳酸盐岩透镜体，煤层
湖坡相（中心区域）
沼泽相、河流相（边缘区域）

分选性差，成分成熟度低

近源搬运

Fig. 5. Cross section of the Tunka depression. 1 = upper series of sediments, 2 = lower series of sediments, 3 = volcanics, 4 = faults, 5 = crystalline basement, 6 = boreholes.

裂谷沉积地层：发育的两个阶段

晚上新世（3~4Ma）~至今

粗粒碎屑沉积
+ 洪积相、滑坡相、冰川相

分选性更差

High Relief, 迅速抬升

渐新世（30~35Ma）~晚上新世（3~4Ma）

细粒碎屑沉积，碳酸盐岩透镜体，煤层
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分选性差，成分成熟度低

近源搬运

Low Relief, 缓慢抬升

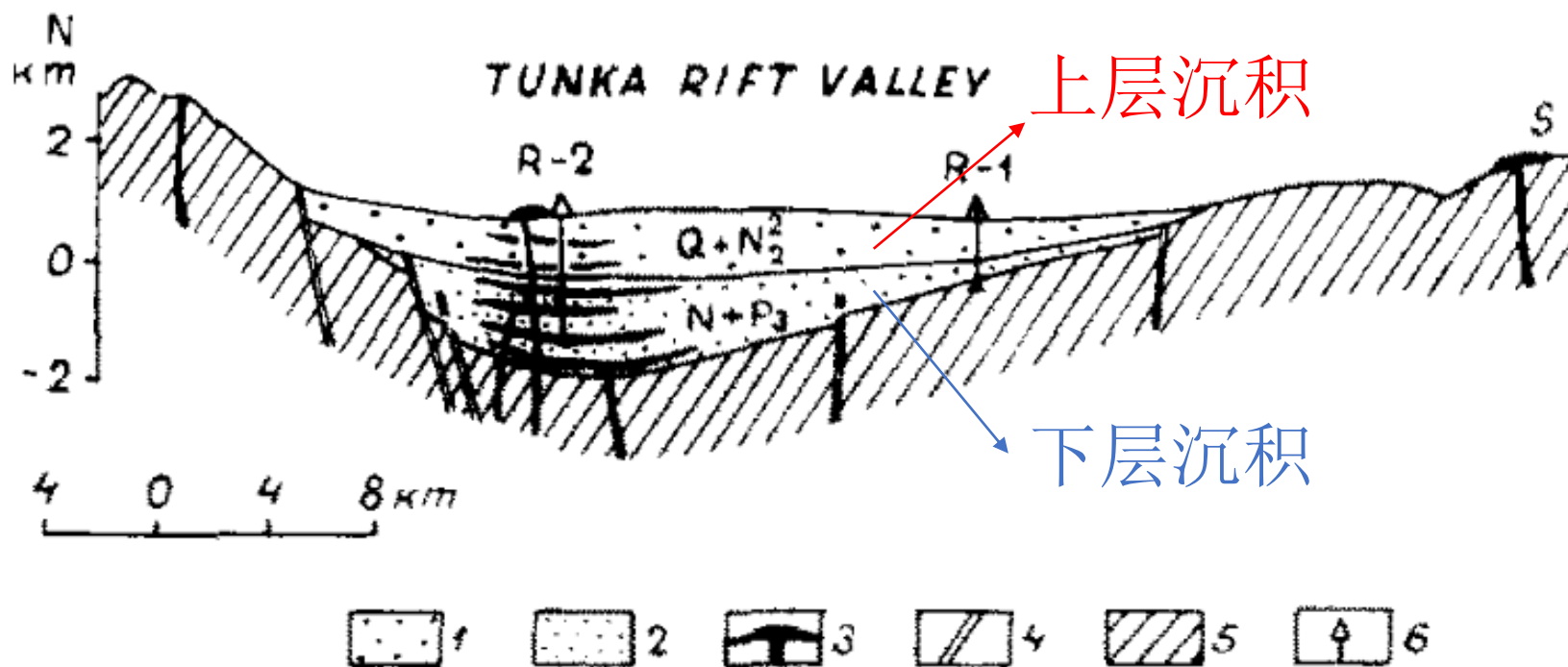
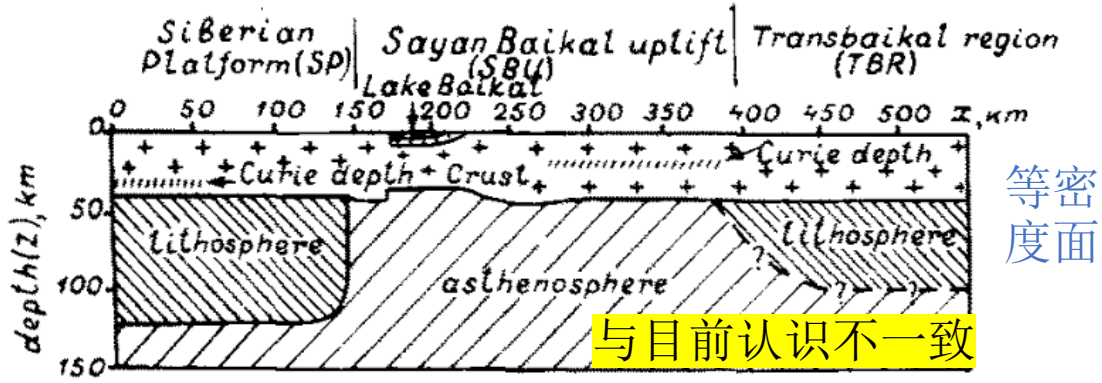
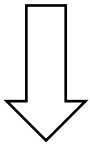


Fig. 5. Cross section of the Tunka depression. 1 = upper series of sediments, 2 = lower series of sediments, 3 = volcanics, 4 = faults, 5 = crystalline basement, 6 = boreholes.

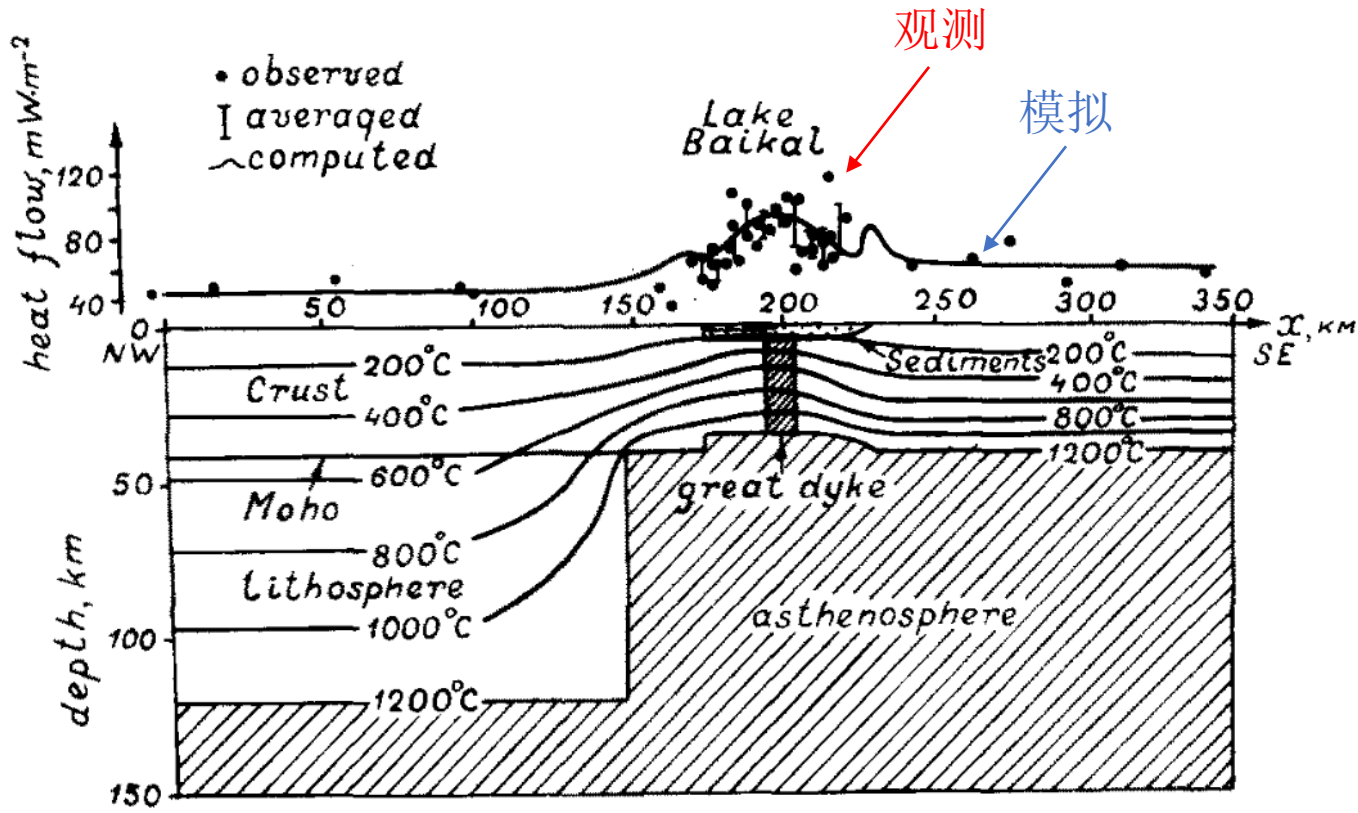
两个阶段解释：主动裂谷端元

见黑板手绘

30~35Ma——3~4Ma

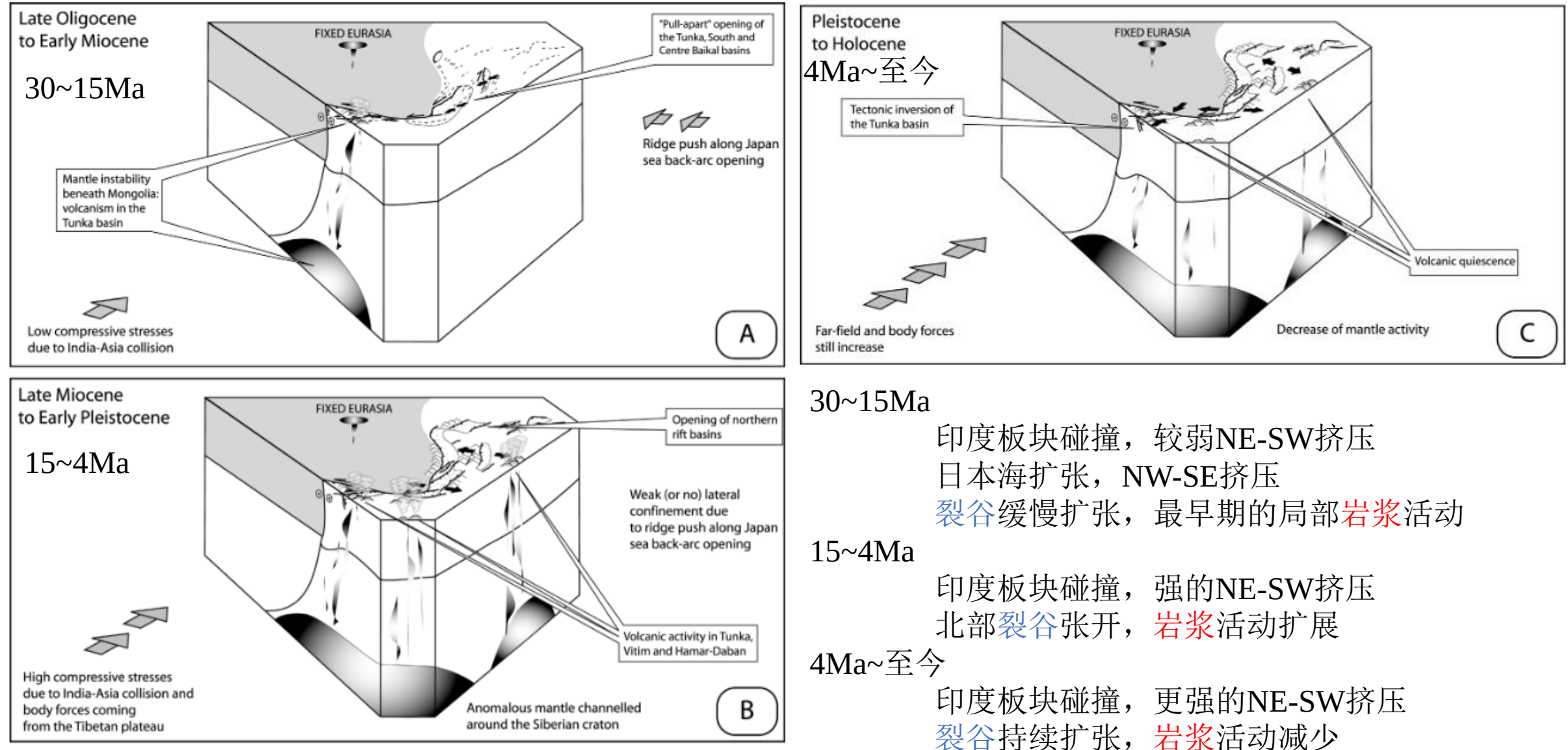


3~4Ma——现在



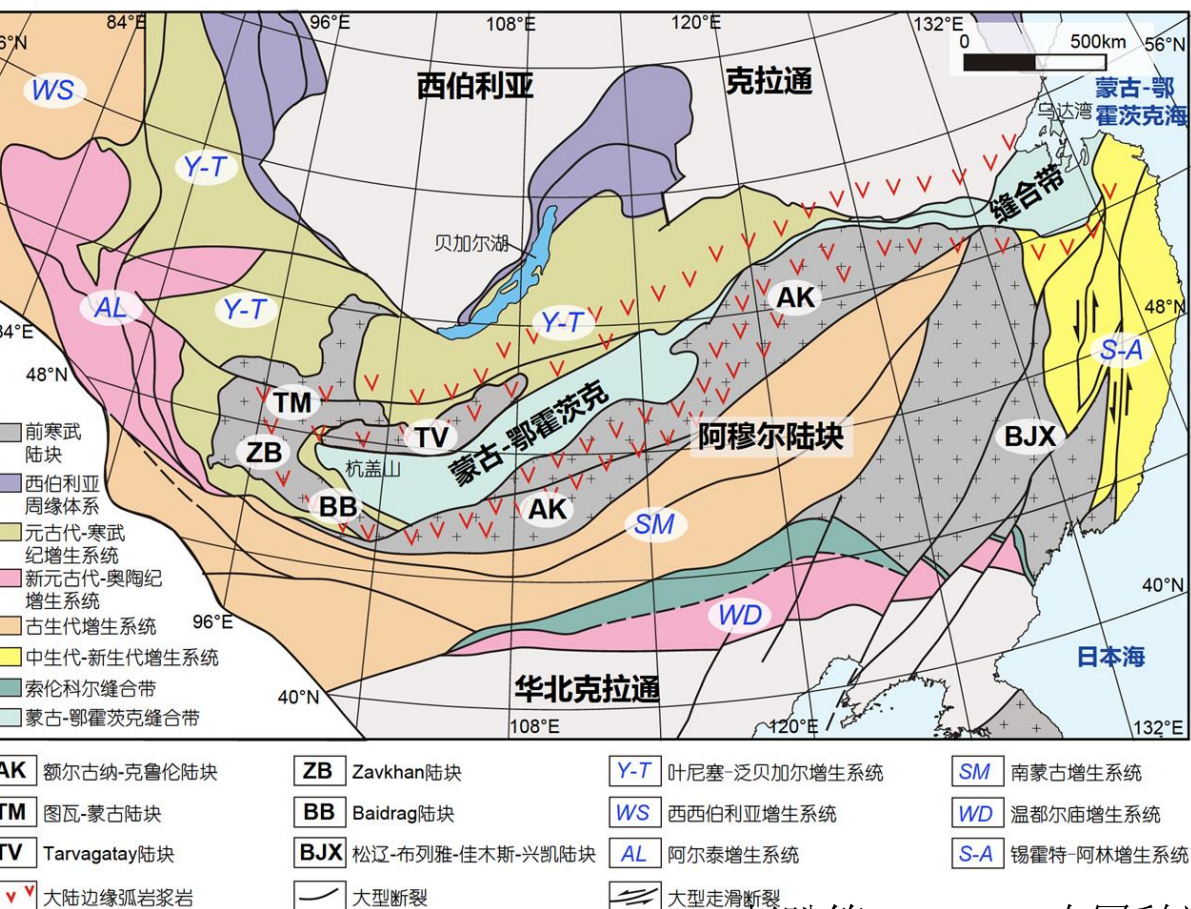
二维数值模拟：热流场
3~4Ma，地壳下出现大规模软流圈

两个阶段解释：被动裂谷端元



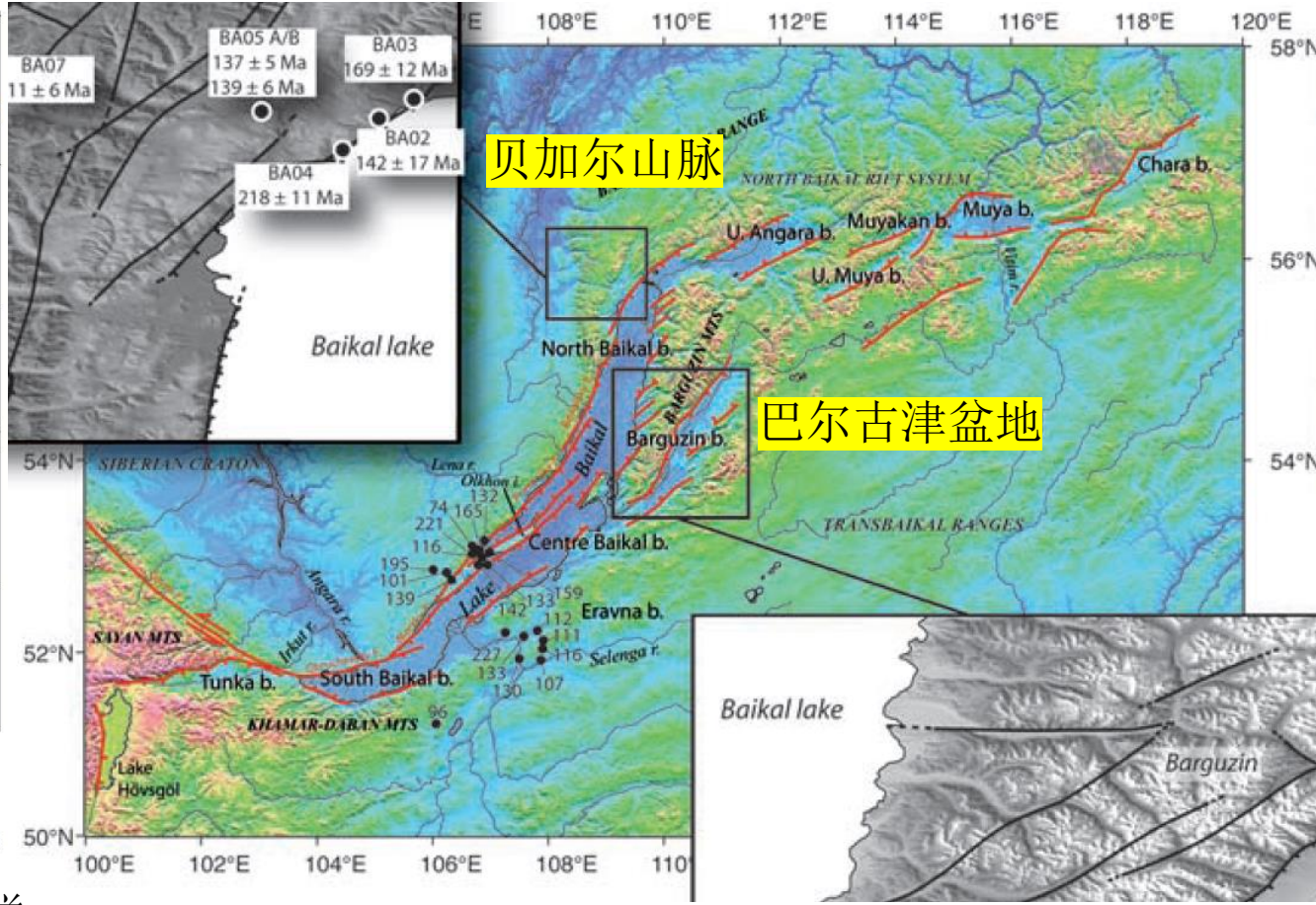
新的观点

AFTA热年代学：裂谷隆升时间



赵盼等, 2023, 中国科学

早白垩世 西伯利亚克拉通与蒙古-鄂霍茨克板块 陆陆碰撞
一般认为: 虽有远端效应, 贝加尔湖整体稳定, 夷平化



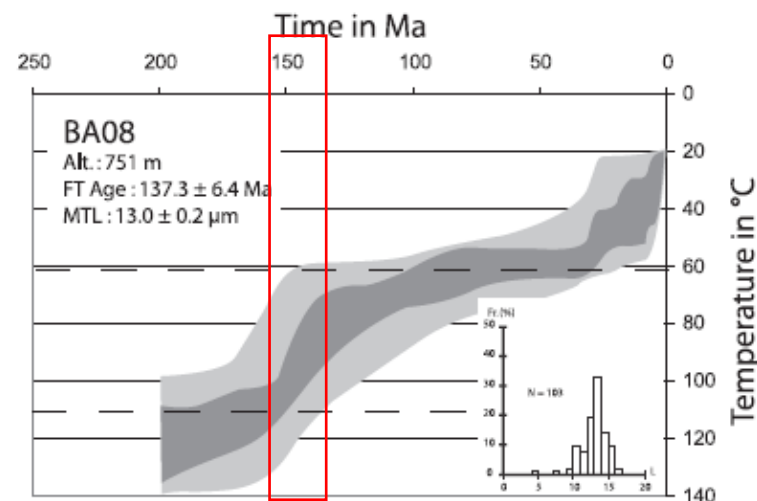
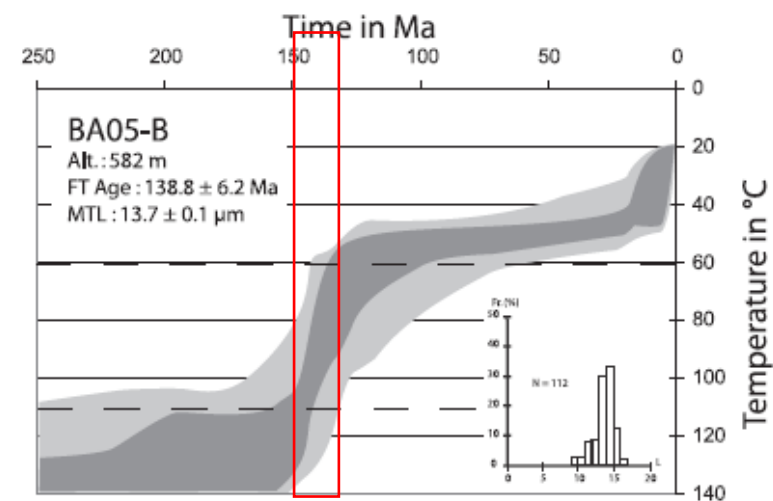
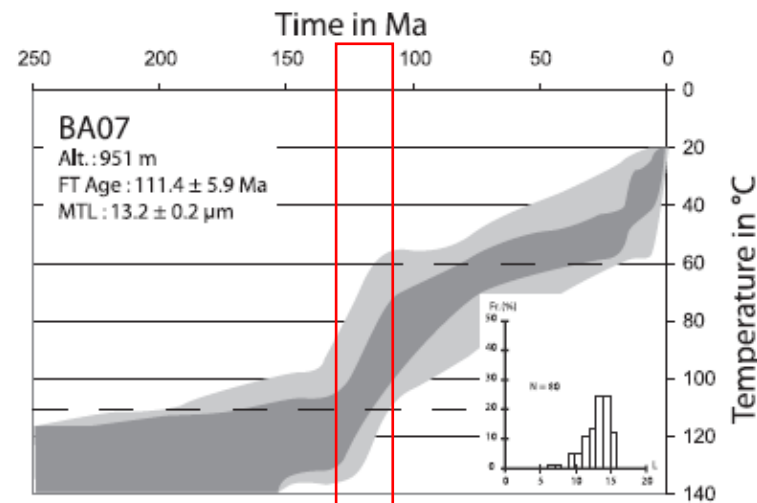
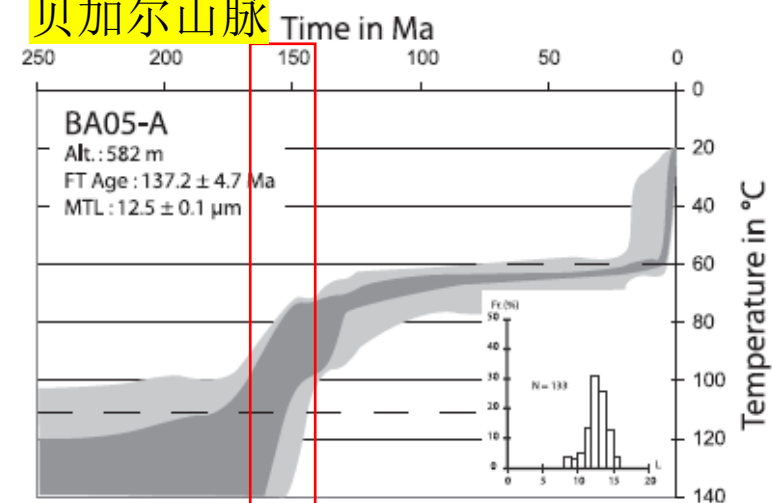
Jolivet et al., 2009, Tectonics

新的观点

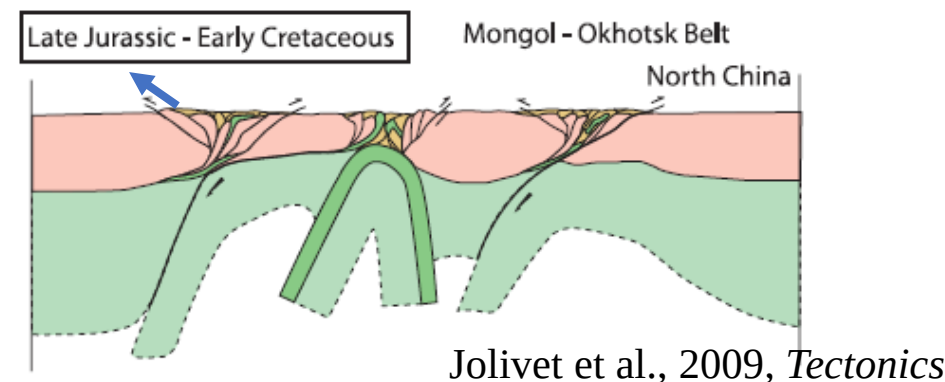
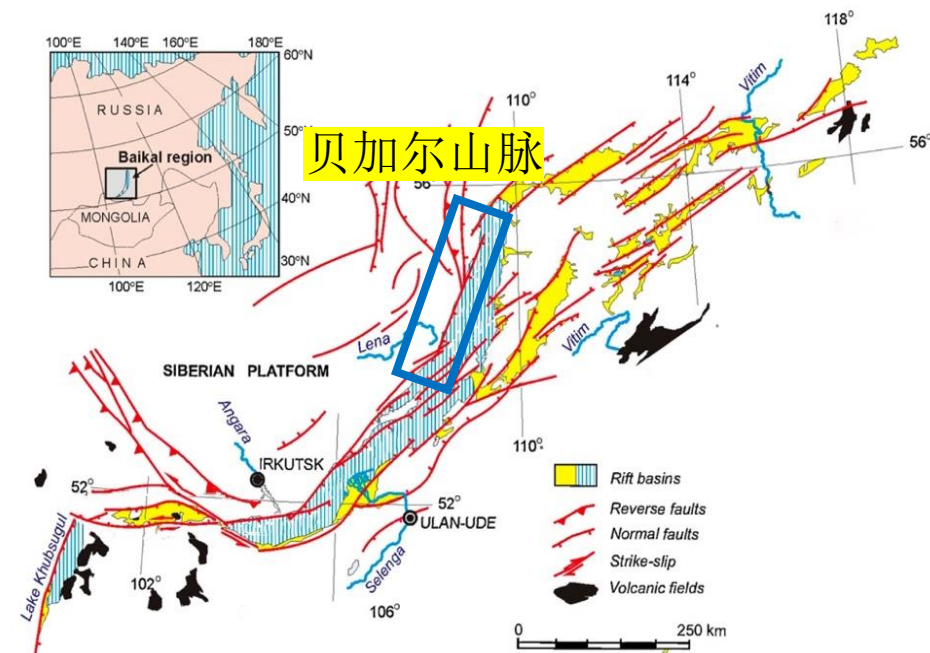
AFTA热年代学：裂谷隆升时间

稳定的夷平面？断层活动加速剥蚀！
剥蚀产物？？？被河流搬运至其余盆地？

贝加尔山脉



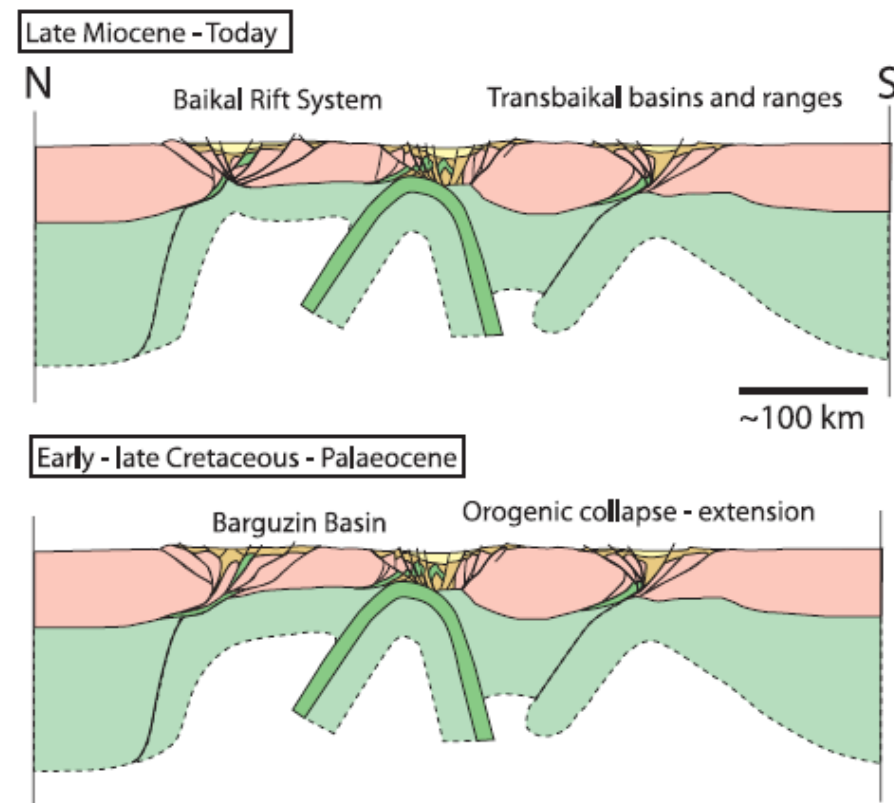
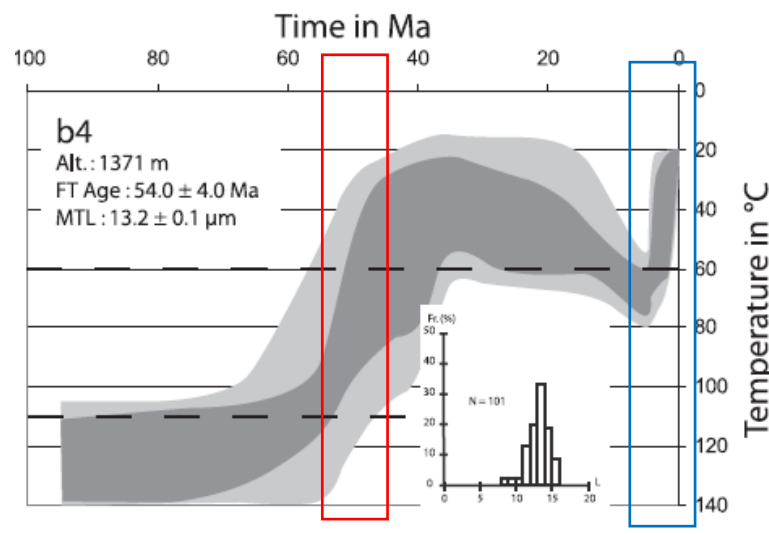
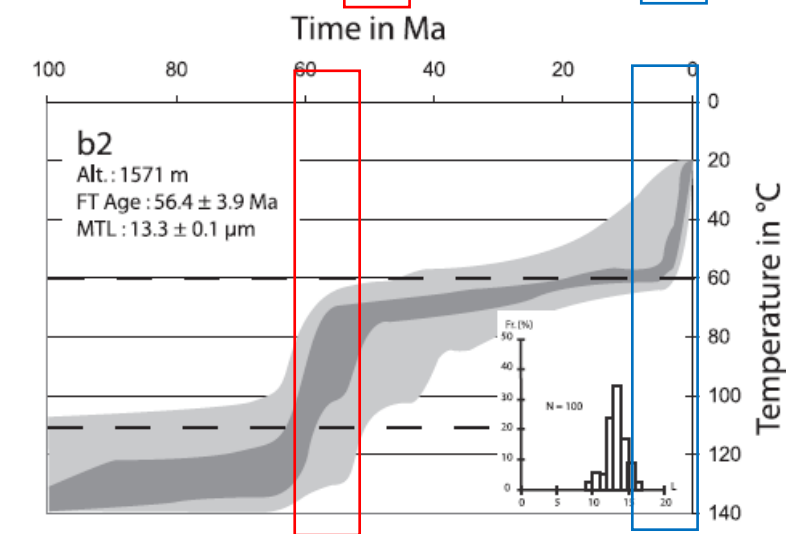
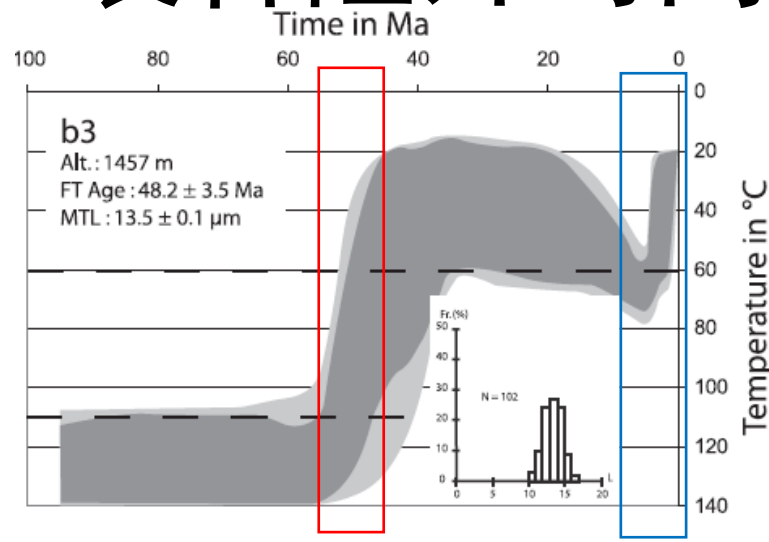
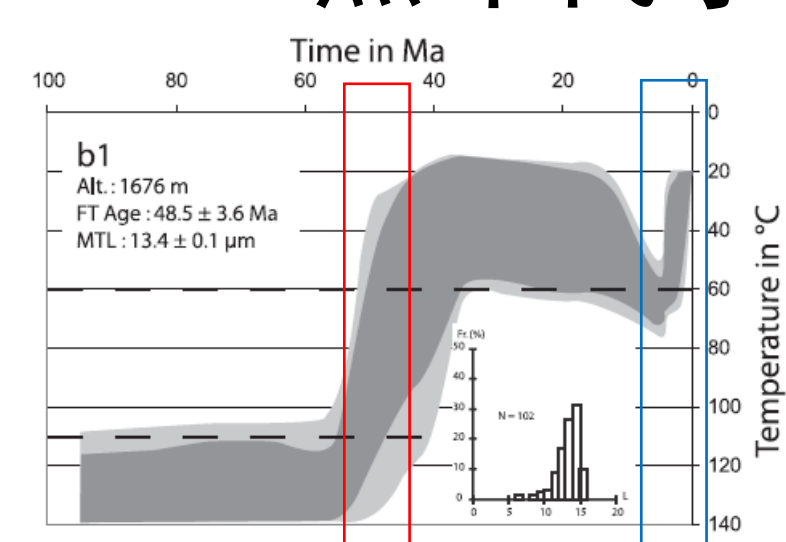
~150Ma 冷却事件 快速隆升 ➡ 断层活动



Jolivet et al., 2009, *Tectonics*

新的观点

AFTA热年代学：裂谷隆升时间



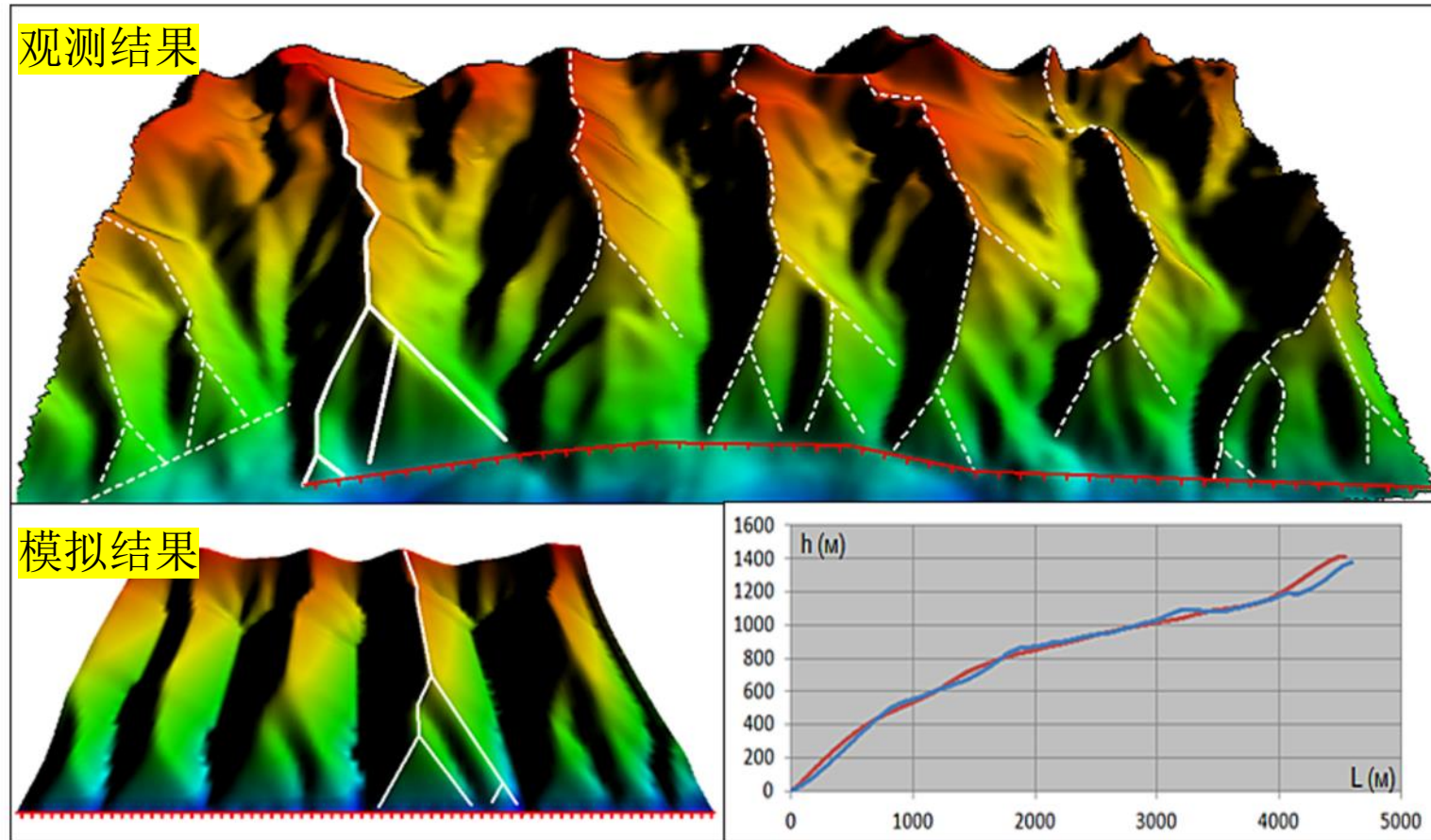
早于部分正断层活动（印亚碰撞）时间
早于幔源岩浆活动（软流圈底辟）时间
以上不是裂谷启动的原因！

太平洋俯冲的弧后拉张？
蒙古-鄂霍茨克造山带的造山后垮塌？

65~50Ma 冷却事件 快速隆升 $\Rightarrow$ 65Ma开始的exhumation $\Rightarrow$ 65Ma 晚白垩世 开始的局部地区裂谷发育

数值模拟：裂谷肩隆升速率

巴尔古津盆地的裂谷肩



断层三角面：快速抬升-剥蚀稳定的产物
➤ 断层三角面的出现—抬升的地貌指示
[形态最“完整”，无次级叠加]

地貌：断层三角面形态

观测结果

内动力：构造抬升速率

外动力：地表过程
气候、滑坡、流水

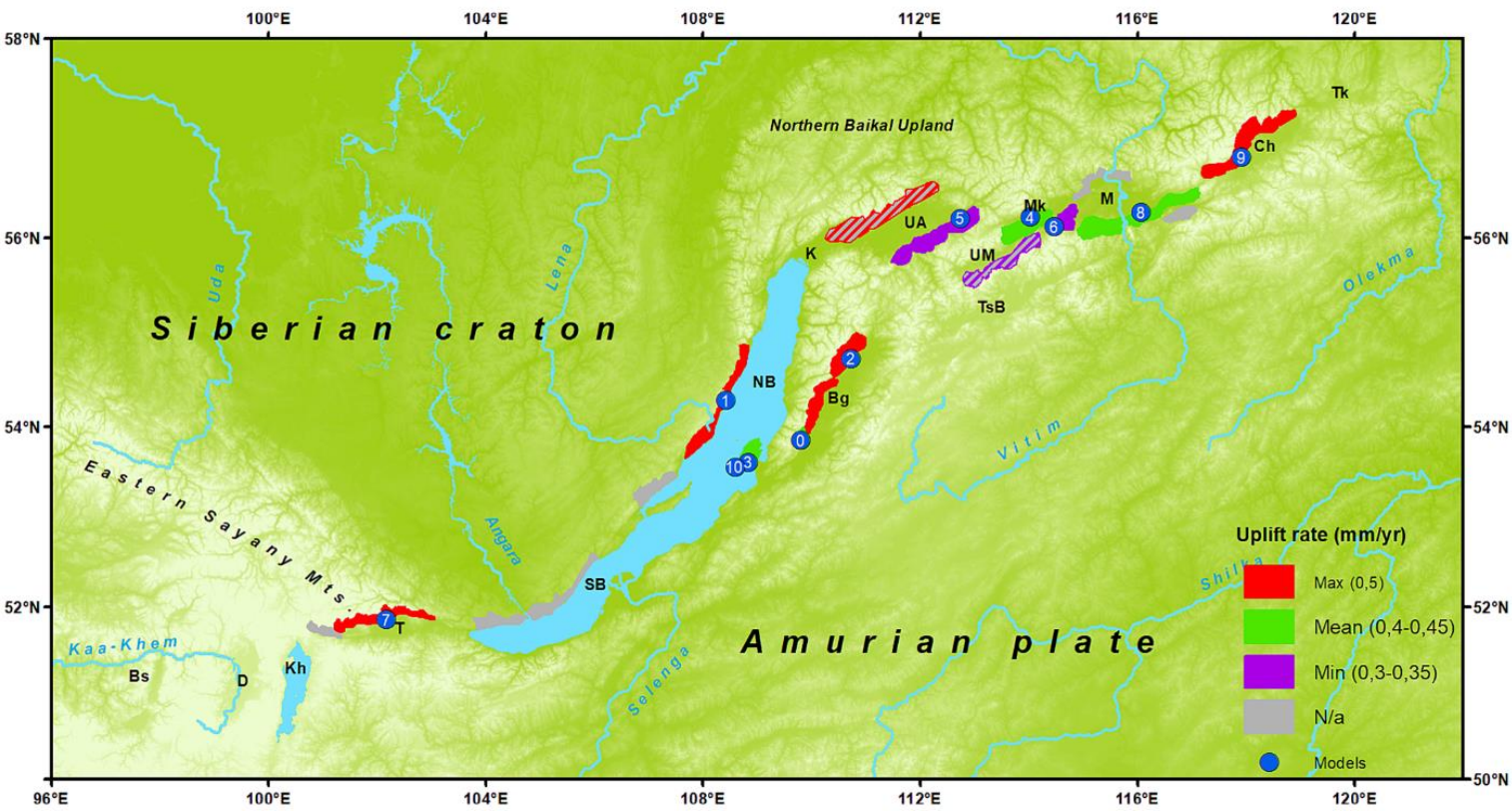
低温热年代学结果

from Jolivet et al., 2009, *Tectonics*

反演求解

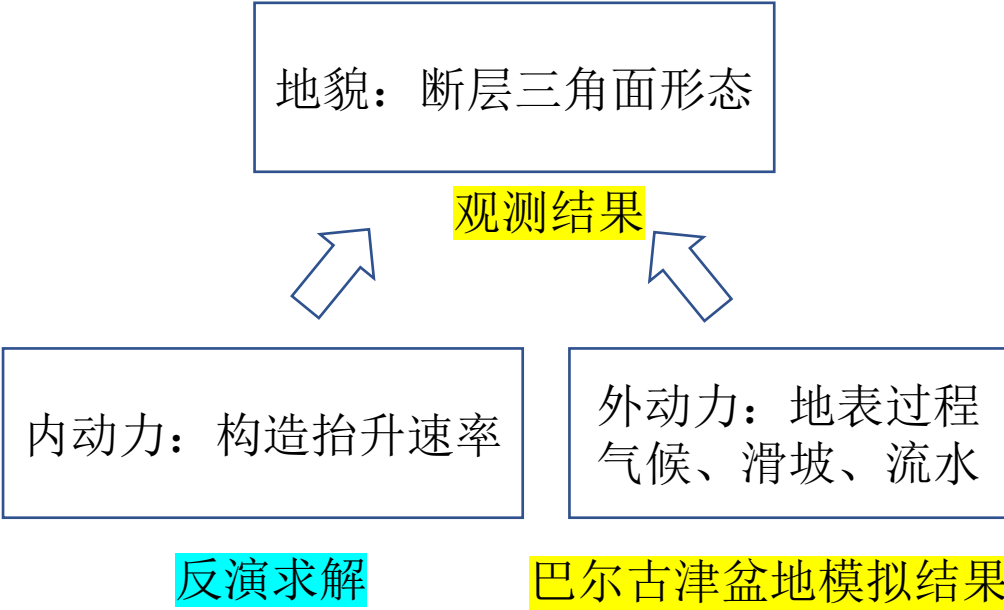
数值模拟：裂谷肩隆升速率

其余盆地的裂谷肩



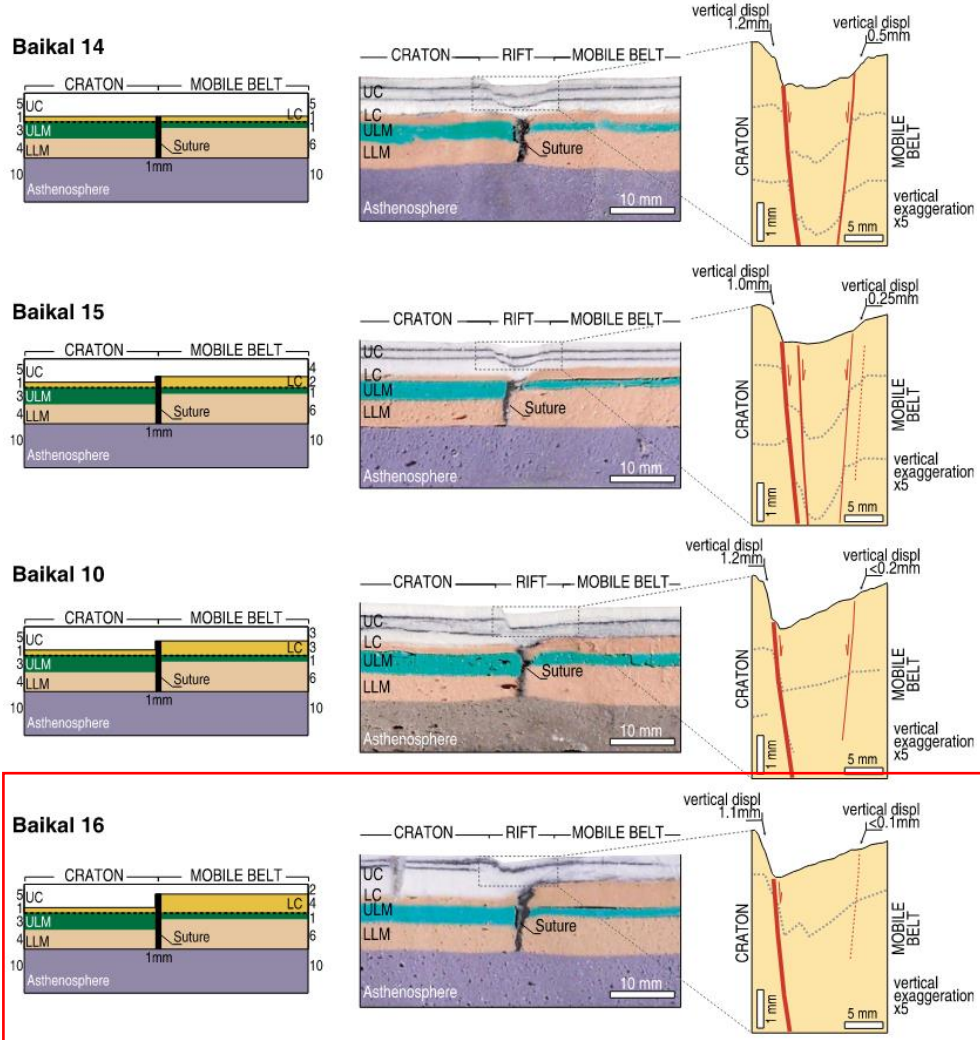
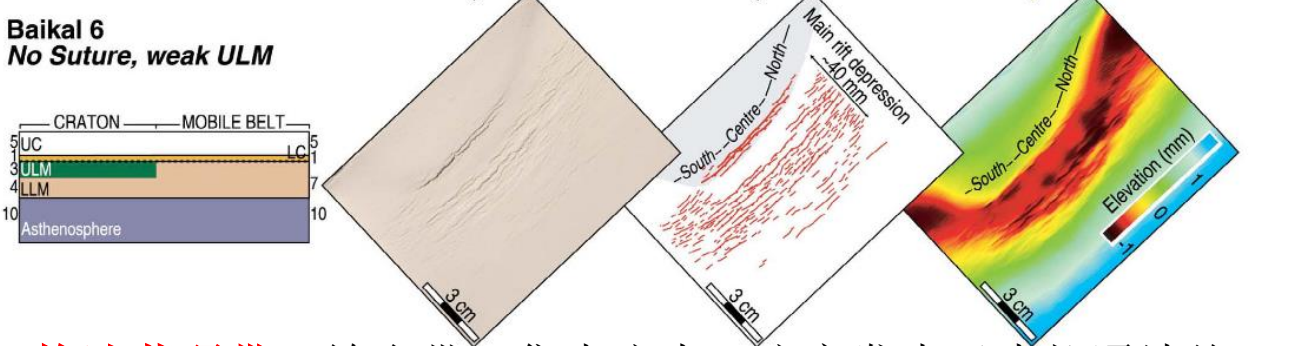
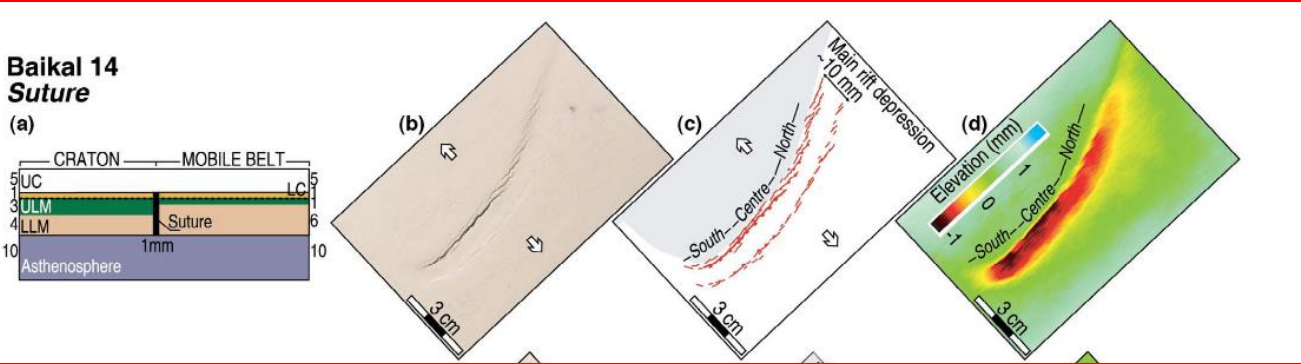
裂谷肩：0.3~0.5mm/yr的隆升速率

断层三角面：快速抬升-剥蚀稳定的产物
➤ 断层三角面的出现—抬升的地貌指示
[形态最“完整”，无次级叠加]



物理模拟：裂谷形态主控因素

裂谷很窄？裂谷两侧不对称？

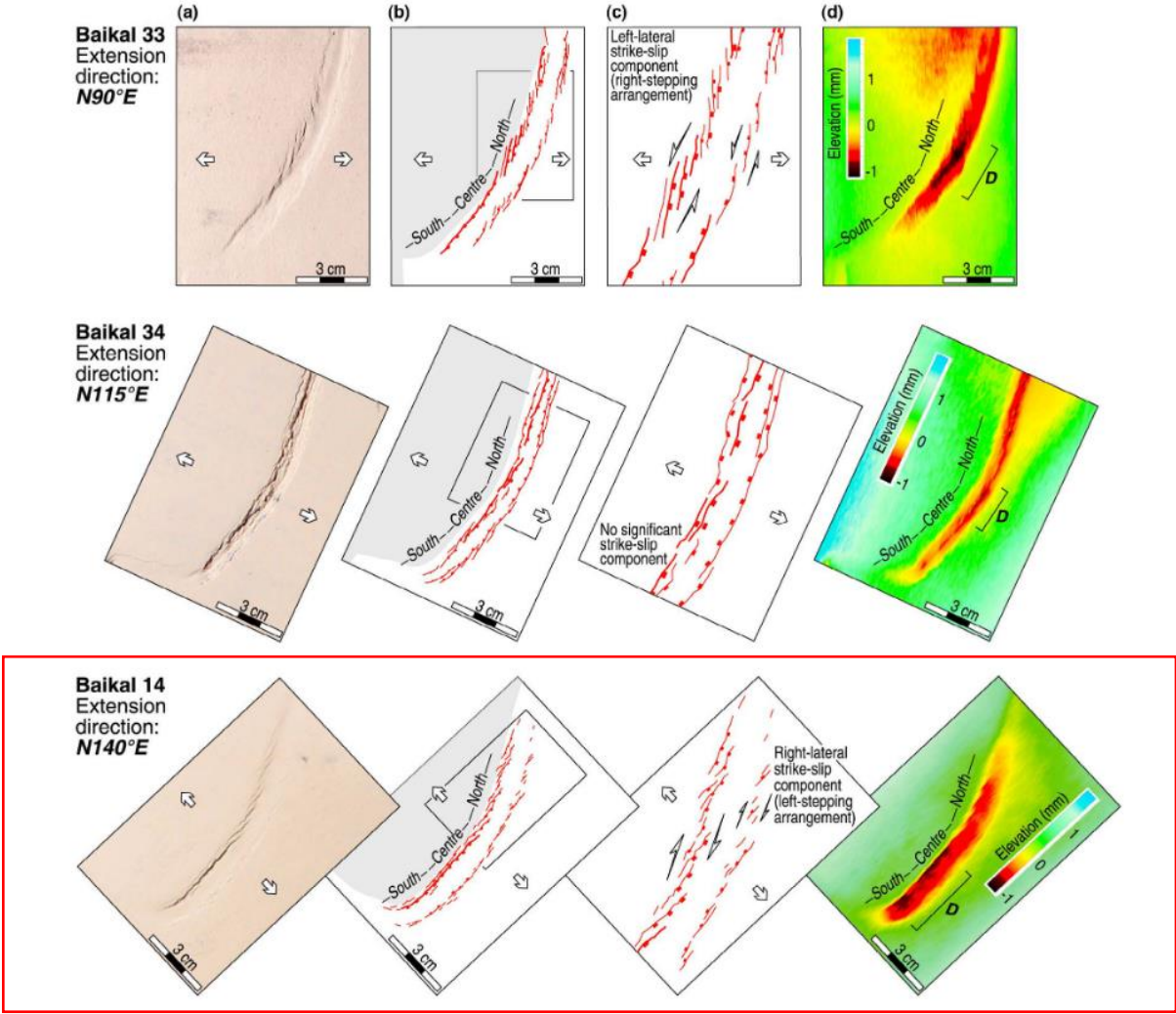


构造薄弱带（缝合带）集中应力，应变发生于克拉通边缘

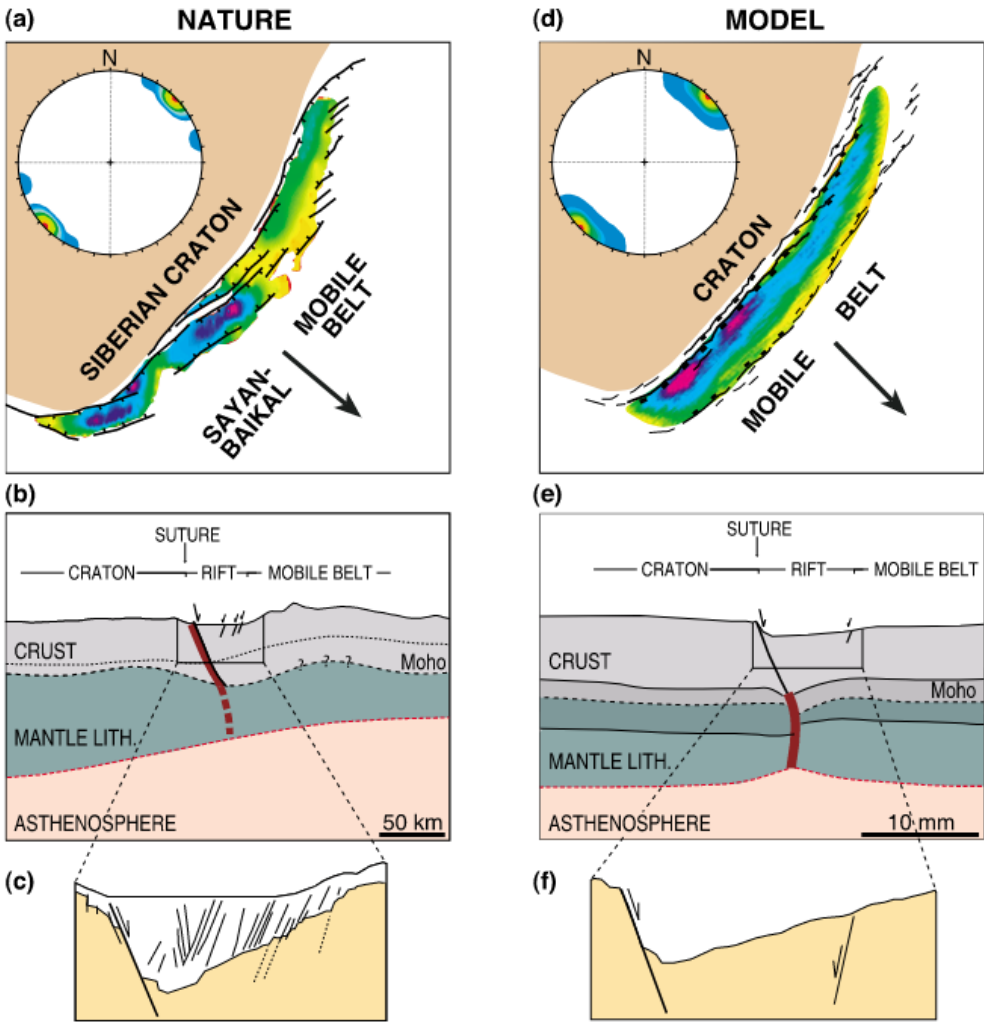
岩石圈流变性差异-脆韧转换带厚度差异-裂谷两侧不对称

物理模拟：裂谷形态主控因素

裂谷沿线不对称？

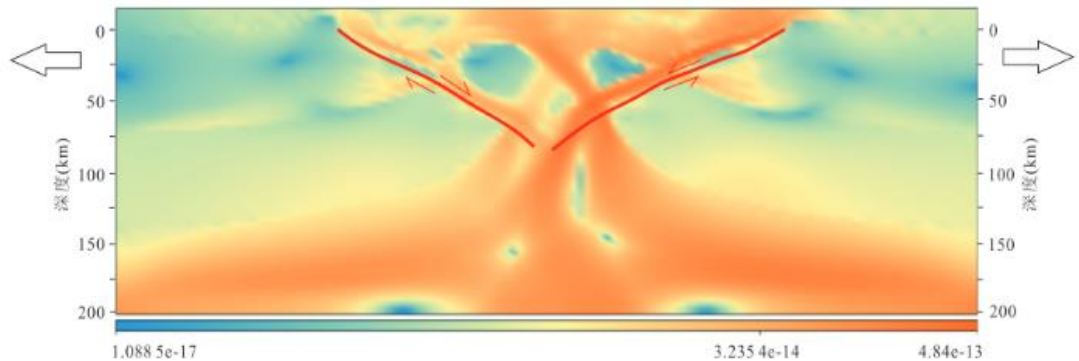


裂谷的运动方向-裂谷沿线的发育情况

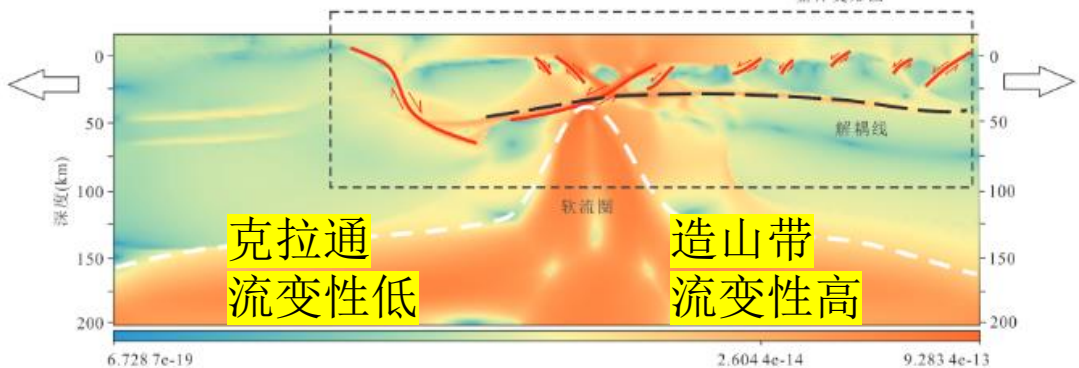


N140° E的模拟最符合实际

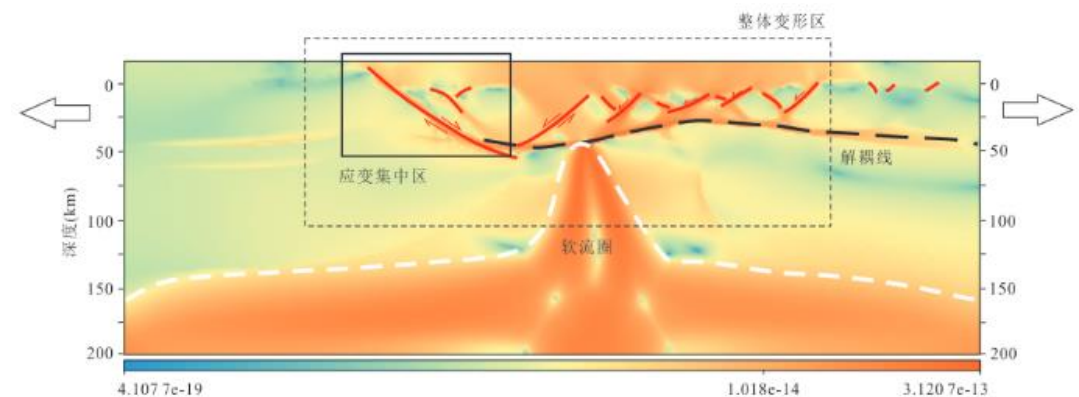
数值模拟：裂谷形态主控因素



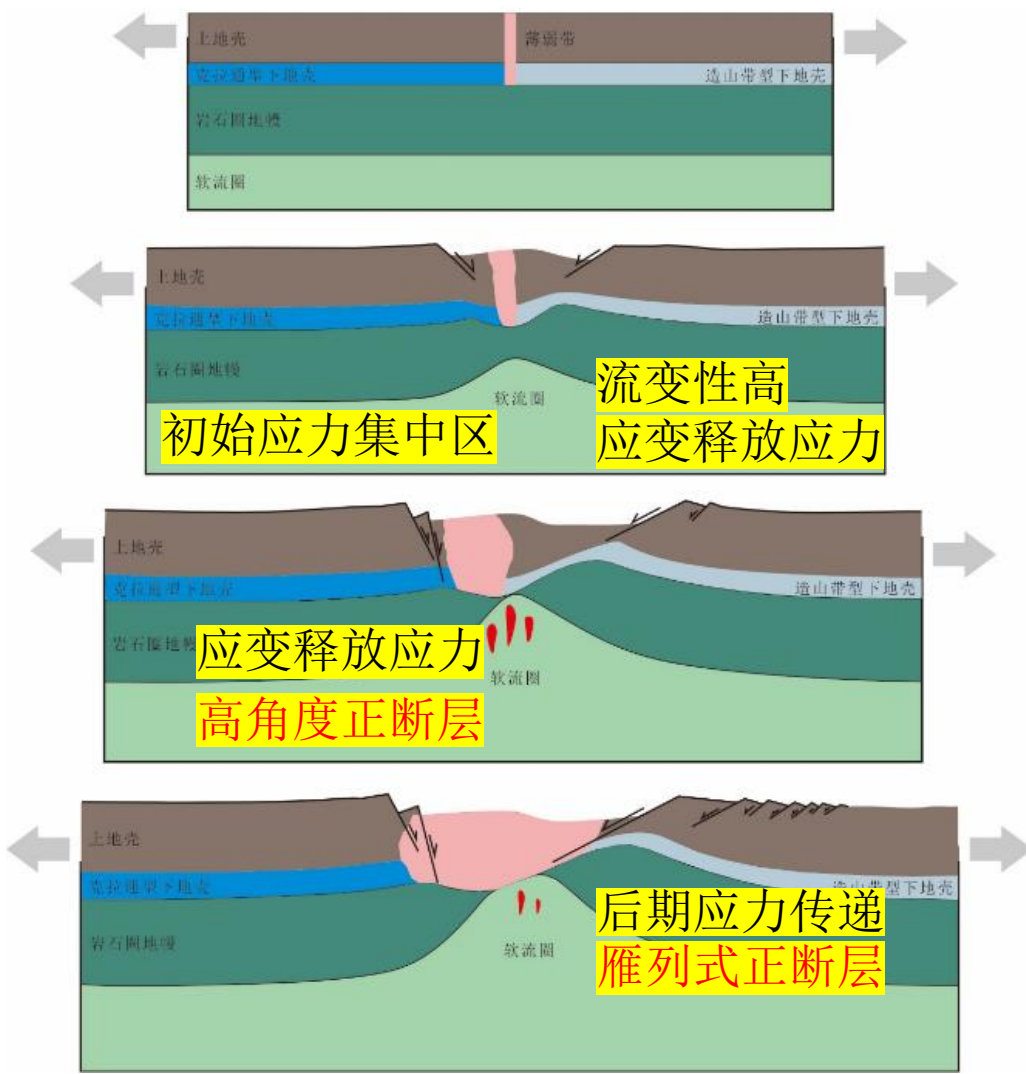
无薄弱带
无造山带
对称裂谷



有薄弱带
无造山带
不对称裂谷
不局域应变



有薄弱带
有造山带
不对称裂谷
局域应变



谢谢！

参考文献（按出现顺序排序）

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